

Supplementary Material

This supplementary material presents additional results associated with the proposed QGG distribution and the QR-QGG regression model. First, plots of the density and survival functions of the QGG distribution are presented for all quantile levels considered, namely, $\tau \in \{0.10, 0.50, 0.90\}$. These plots illustrate the effects of different parameter settings on the shape of the distribution and the behavior of the survival function.

Next, a Monte Carlo simulation study is presented to investigate the finite-sample performance of the parameter estimators for the proposed QGG distribution and the QR-QGG regression model. For each scenario, 10,000 Monte Carlo replications were considered, with sample sizes $n \in \{100, 200, 400\}$. The performance of the maximum likelihood estimators was assessed using the mean, bias, standard deviation (SD), and mean squared error (MSE). In addition, confidence intervals constructed with a 95% confidence level were evaluated through their coverage probability (CP).

The Monte Carlo study was implemented in the R software environment using computational routines developed by the authors. Parameter estimation was performed by maximum likelihood, with numerical optimization based on the Nelder–Mead algorithm. To reduce numerical instability and facilitate convergence of the iterative procedure, a preliminary step was adopted to obtain initial values.

In this step, preliminary estimates were obtained by minimizing the sum of squares of the score vector. This procedure used the `nls.lm()` function from the `minpack.lm` package, which employs a modification of the Levenberg–Marquardt algorithm to minimize the sum of squares of the vector returned by the objective function. The vector of initial values was defined as follows: for the parameter μ_τ , the empirical quantile corresponding to level τ was used; for the parameters σ and ν , random initial values were generated from a uniform distribution on the interval $(0, 2)$.

Using these preliminary values, the final parameter estimates were obtained with the Nelder–Mead algorithm. During the simulation study, the nonconvergence rate ranged from 10% to 15% for the QGG distribution, whereas for the QR-QGG regression model it was approximately 10%.

1. Graphical behavior of the QGG distribution— Density function

This section presents plots of the density function of the $\text{QGG}(\mu_\tau, \sigma, \nu)$ distribution for different settings of the parameters μ_τ , σ , and ν , considering the quantile levels $\tau \in \{0.10, 0.50, 0.90\}$. These plots show how changes in the parameters affect the location, concentration, and shape of the density over the positive support.

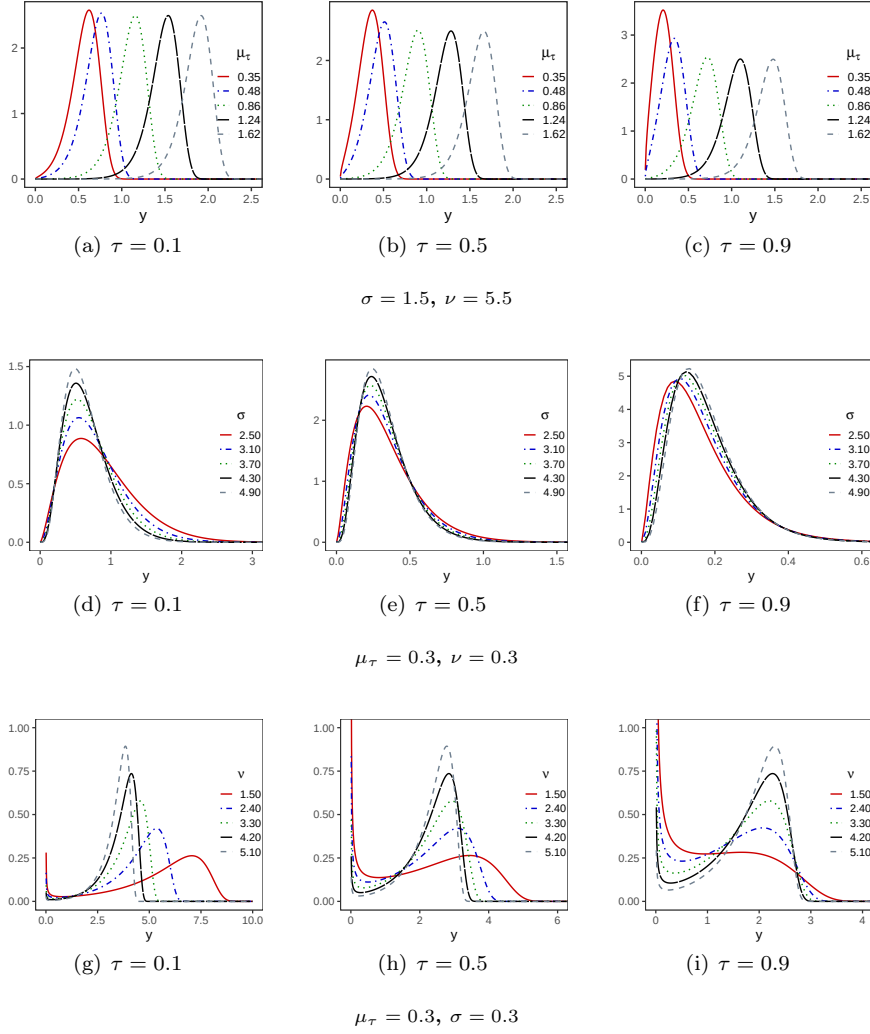


FIGURE 1: Probability density functions of the $\text{QGG}(\mu_\tau, \sigma, \nu)$ distribution for $\tau = (0.1, 0.5, 0.9)$, considering variations in μ_τ , σ , and ν .

2. Graphical behavior of the QGG distribution—Survival function

The survival function of the $\text{QGG}(\mu_\tau, \sigma, \nu)$ distribution is analyzed for the quantile levels $\tau \in \{0.10, 0.50, 0.90\}$. The plots presented in this section describe the probability that the random variable takes values greater than y , considering different combinations of the parameters μ_τ , σ , and ν . Thus, the plots show how

these combinations influence the decay pattern of the survival function over the positive support.

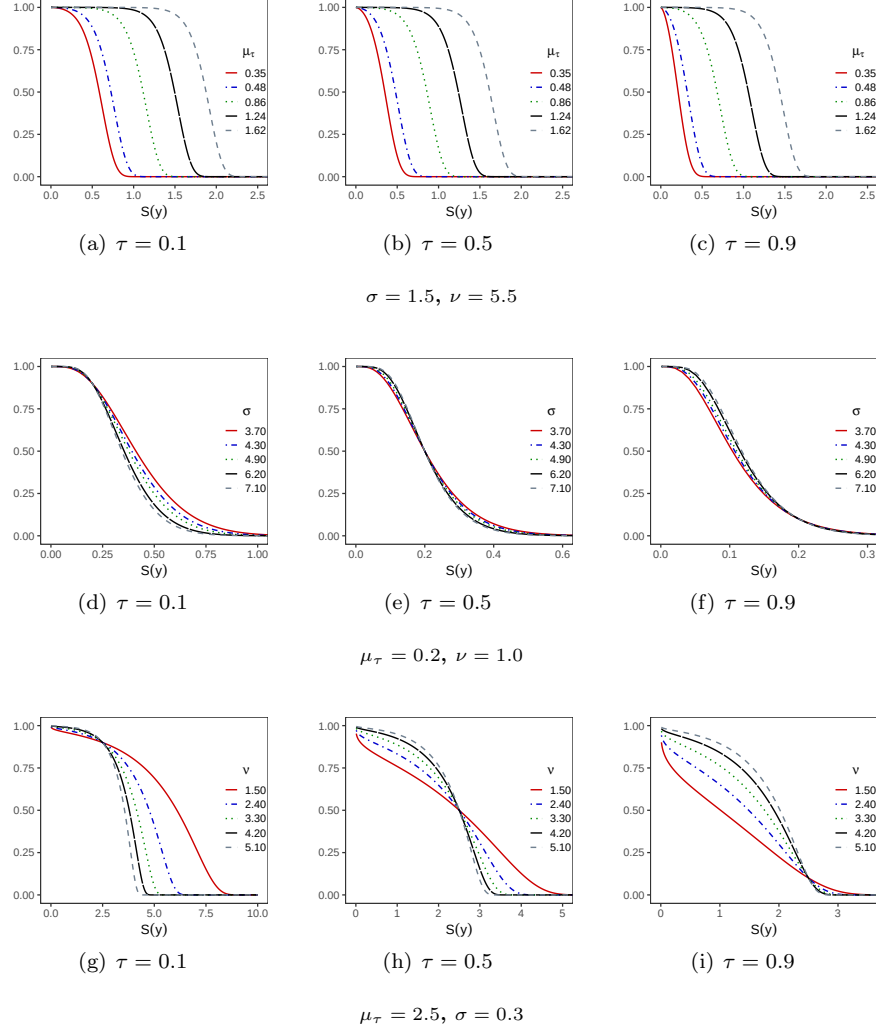


FIGURE 2: Survival functions of the $QGG(\mu_\tau, \sigma, \nu)$ distribution for $\tau = (0.1, 0.5, 0.9)$, considering variations in μ_τ , σ , and ν .

3. Monte Carlo simulations for the QGG distribution

This section presents the results of the Monte Carlo simulations conducted to evaluate the finite-sample performance of the maximum likelihood estimators of the QGG distribution parameters.

3.1. Scenario 1— $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$

TABLE 1: Monte Carlo simulation results for the QGG(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	2.8715	1.6117	0.5396	2.8462	1.5486	0.5263	2.8360	1.5392	0.5169
	200	2.8649	1.5408	0.5217	2.8484	1.5224	0.5118	2.8444	1.5205	0.5083
	400	2.8559	1.5219	0.5085	2.8491	1.5105	0.5063	2.8469	1.5079	0.5043
	800	2.8519	1.5123	0.5033	2.8496	1.5041	0.5036	2.8479	1.5037	0.5023
Bias	100	0.0215	0.1117	0.0396	-0.0038	0.0486	0.0263	-0.0140	0.0392	0.0169
	200	0.0149	0.0408	0.0217	-0.0016	0.0224	0.0118	-0.0056	0.0205	0.0083
	400	0.0059	0.0219	0.0085	-0.0009	0.0105	0.0063	-0.0031	0.0079	0.0043
	800	0.0019	0.0123	0.0033	-0.0004	0.0041	0.0036	-0.0021	0.0037	0.0023
SD	100	0.3152	0.4951	0.2289	0.1294	0.3434	0.1521	0.0922	0.3009	0.1162
	200	0.2249	0.3043	0.1427	0.0918	0.2261	0.0989	0.0649	0.2064	0.0781
	400	0.1598	0.1989	0.0938	0.0657	0.1548	0.0664	0.0460	0.1436	0.0538
	800	0.1137	0.1376	0.0640	0.0463	0.1087	0.0493	0.0324	0.0995	0.0375
MSE	100	0.0998	0.2576	0.0539	0.0168	0.1203	0.0238	0.0087	0.0921	0.0138
	200	0.0508	0.0943	0.0208	0.0084	0.0516	0.0099	0.0042	0.0430	0.0062
	400	0.0256	0.0400	0.0089	0.0043	0.0241	0.0045	0.0021	0.0207	0.0029
	800	0.0129	0.0191	0.0041	0.0021	0.0118	0.0024	0.0011	0.0099	0.0014
CP	100	0.8934	0.8708	0.8987	0.9387	0.9248	0.9445	0.9383	0.9493	0.9511
	200	0.9061	0.9034	0.9247	0.9430	0.9339	0.9479	0.9421	0.9457	0.9491
	400	0.9149	0.9203	0.9305	0.9398	0.9375	0.9437	0.9467	0.9436	0.9450
	800	0.9140	0.9245	0.9304	0.9459	0.9312	0.9393	0.9478	0.9406	0.9441

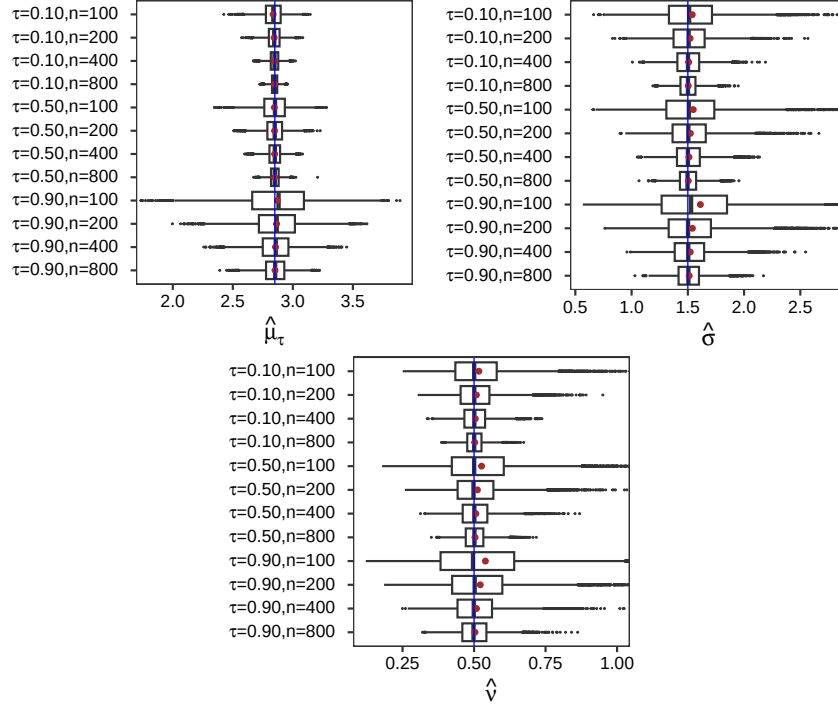


FIGURE 3: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $QGG(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$.
Legend: —: true parameter value; •: mean of the 10,000 replications.

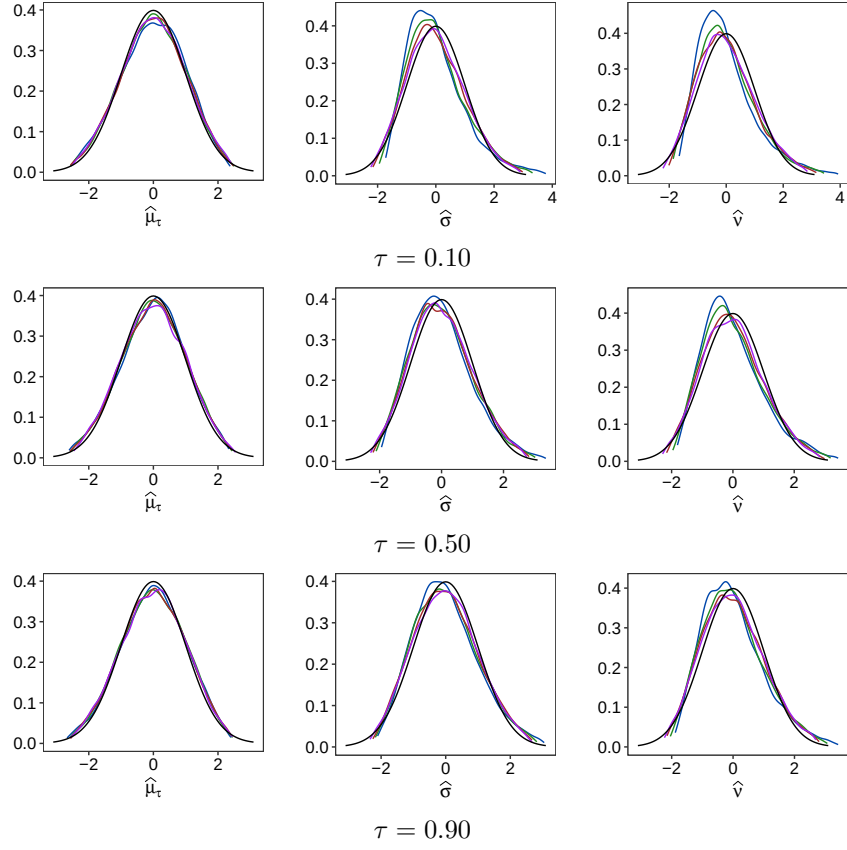
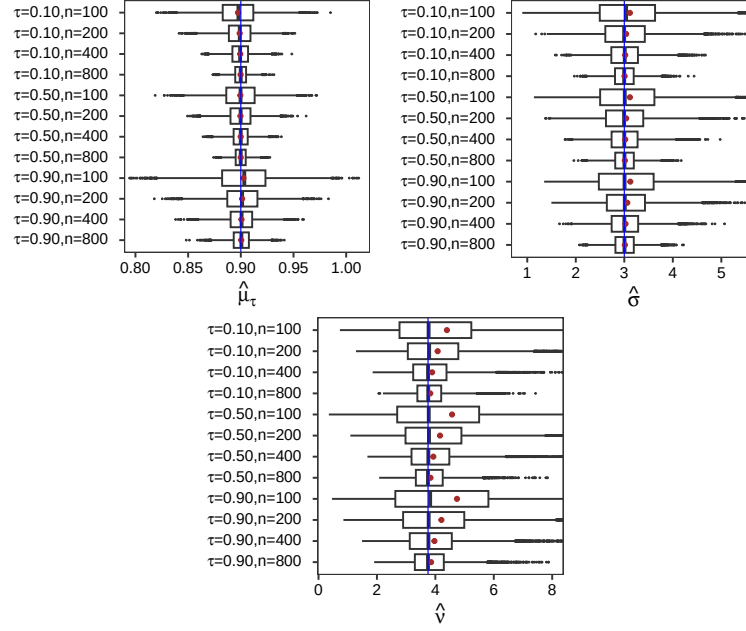


FIGURE 4: Sampling densities of the maximum likelihood estimators of the parameters of the $QGG(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$, $\tau \in \{0.10, 0.50, 0.90\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.2. Scenario 2— $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$

TABLE 2: Monte Carlo simulation results for the QGG(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 0.9$, $\sigma = 3$, and $\nu = 3.75$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	0.9026	3.1241	4.7409	0.8996	3.1161	4.5718	0.8974	3.1150	4.3958
	200	0.9013	3.0617	4.2079	0.8997	3.0388	4.1598	0.8989	3.0381	4.0796
	400	0.9006	3.0252	3.9689	0.8999	3.0189	3.9312	0.8995	3.0193	3.8895
	800	0.9004	3.0118	3.8555	0.8999	3.0118	3.8352	0.8998	3.0068	3.8258
Bias	100	0.0026	0.1241	0.9909	-0.0004	0.1161	0.8218	-0.0026	0.1150	0.6458
	200	0.0013	0.0617	0.4579	-0.0003	0.0388	0.4098	-0.0011	0.0381	0.3296
	400	0.0006	0.0252	0.2189	-0.0001	0.0189	0.1812	-0.0005	0.0193	0.1395
	800	0.0004	0.0118	0.1055	-0.0001	0.0118	0.0852	-0.0002	0.0068	0.0758
SD	100	0.0305	0.9265	3.0915	0.0200	0.9016	2.7942	0.0217	0.8954	2.4119
	200	0.0214	0.6069	1.9287	0.0141	0.5980	1.7409	0.0154	0.6051	1.4873
	400	0.0155	0.4115	1.2122	0.0098	0.4053	1.0495	0.0109	0.4134	0.9264
	800	0.0108	0.2830	0.7892	0.0071	0.2856	0.7128	0.0077	0.2905	0.6310
MSE	100	0.0009	0.8738	10.5396	0.0004	0.8263	8.4831	0.0005	0.8150	6.2344
	200	0.0005	0.3722	3.9296	0.0002	0.3592	3.1986	0.0002	0.3676	2.3207
	400	0.0002	0.1700	1.5174	0.0001	0.1647	1.1342	0.0001	0.1713	0.8778
	800	0.0001	0.0802	0.6340	0.0000	0.0817	0.5153	0.0001	0.0845	0.4038
CP	100	0.9310	0.9580	0.9007	0.9417	0.9602	0.9144	0.9328	0.9635	0.9217
	200	0.9392	0.9529	0.9254	0.9440	0.9531	0.9308	0.9405	0.9533	0.9367
	400	0.9385	0.9495	0.9378	0.9516	0.9521	0.9476	0.9434	0.9512	0.9499
	800	0.9414	0.9515	0.9479	0.9493	0.9487	0.9465	0.9485	0.9502	0.9482



(b) Scenario 2: $\mu_\tau = 1.86$, $\sigma = 0.93$ and $\nu = 0.42$

FIGURE 5: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $\text{QGG}(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

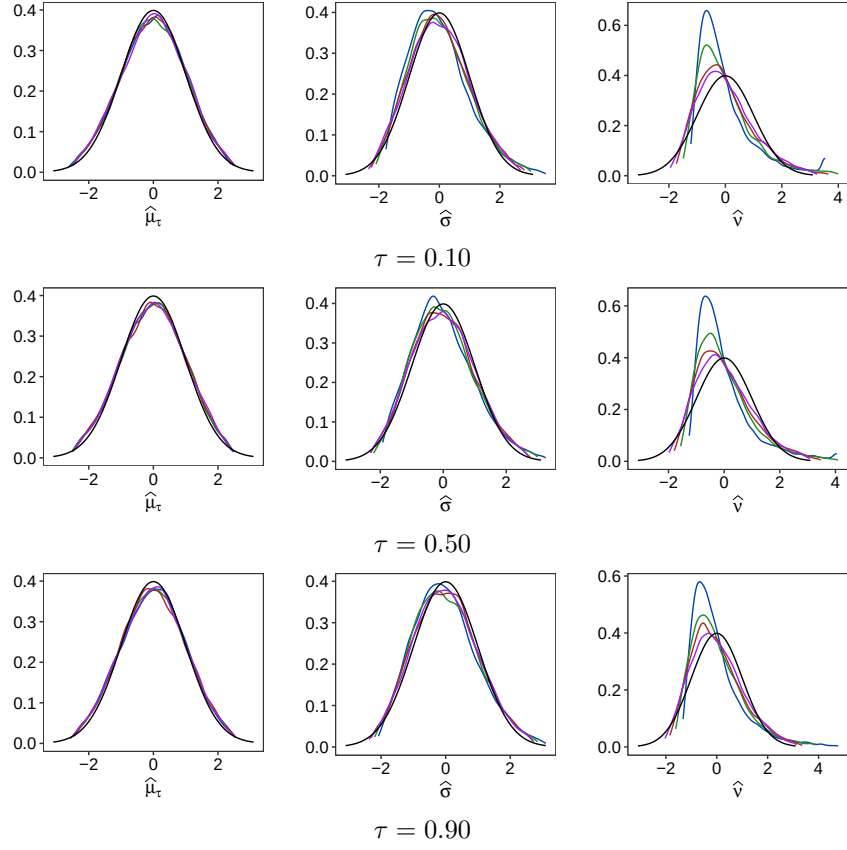


FIGURE 6: Sampling densities of the maximum likelihood estimators of the parameters of the $QGG(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.3. Scenario 3— $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$

TABLE 3: Monte Carlo simulation results for the QGG(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	3.6523	2.5659	2.5595	3.6502	2.5604	2.5477	3.6463	2.5572	2.5345
	200	3.6514	2.4806	2.2089	3.6502	2.4746	2.2244	3.6485	2.4599	2.2440
	400	3.6510	2.4323	2.1004	3.6499	2.4319	2.1036	3.6490	2.4354	2.0898
	800	3.6508	2.4166	2.0487	3.6500	2.4155	2.0480	3.6493	2.4163	2.0454
Bias	100	0.0023	0.1659	0.5595	0.0002	0.1604	0.5477	-0.0037	0.1572	0.5345
	200	0.0014	0.0806	0.2089	0.0002	0.0746	0.2244	-0.0015	0.0599	0.2440
	400	0.0010	0.0323	0.1004	-0.0001	0.0319	0.1036	-0.0010	0.0354	0.0898
	800	0.0008	0.0166	0.0487	0.0000	0.0155	0.0480	-0.0007	0.0163	0.0454
SD	100	0.0611	0.8111	2.0650	0.0360	0.8004	2.0062	0.0345	0.7936	1.9845
	200	0.0436	0.5104	1.0286	0.0251	0.5041	1.0637	0.0246	0.4987	1.0517
	400	0.0310	0.3342	0.6140	0.0179	0.3368	0.6228	0.0174	0.3316	0.5961
	800	0.0217	0.2319	0.4042	0.0126	0.2342	0.4058	0.0122	0.2297	0.3976
MSE	100	0.0037	0.6854	4.5773	0.0013	0.6664	4.3246	0.0012	0.6545	4.2240
	200	0.0019	0.2670	1.1016	0.0006	0.2597	1.1817	0.0006	0.2523	1.1656
	400	0.0010	0.1127	0.3871	0.0003	0.1145	0.3986	0.0003	0.1112	0.3634
	800	0.0005	0.0541	0.1657	0.0002	0.0551	0.1670	0.0001	0.0530	0.1601
CP	100	0.9351	0.9482	0.8890	0.9404	0.9521	0.8887	0.9363	0.9508	0.8891
	200	0.9385	0.9482	0.9143	0.9441	0.9451	0.9183	0.9441	0.9452	0.9218
	400	0.9408	0.9516	0.9382	0.9454	0.9497	0.9327	0.9436	0.9543	0.9357
	800	0.9424	0.9502	0.9434	0.9465	0.9490	0.9403	0.9496	0.9526	0.9447

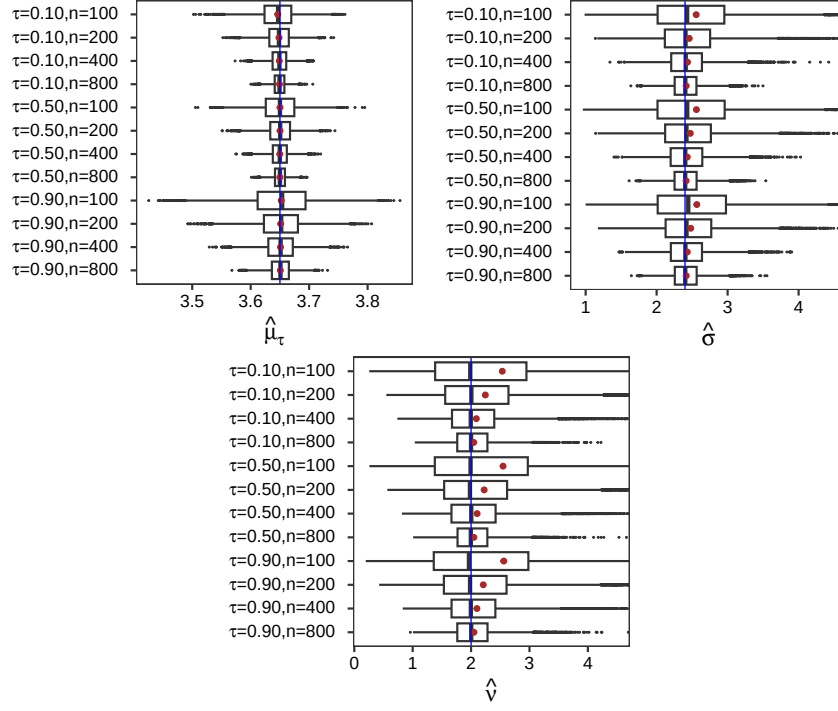


FIGURE 7: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the QGG(μ_τ, σ, ν) distribution, with $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$, considering $n \in \{100, 200, 400\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: --- : true parameter value; $\bullet$: mean of the 10,000 replications.

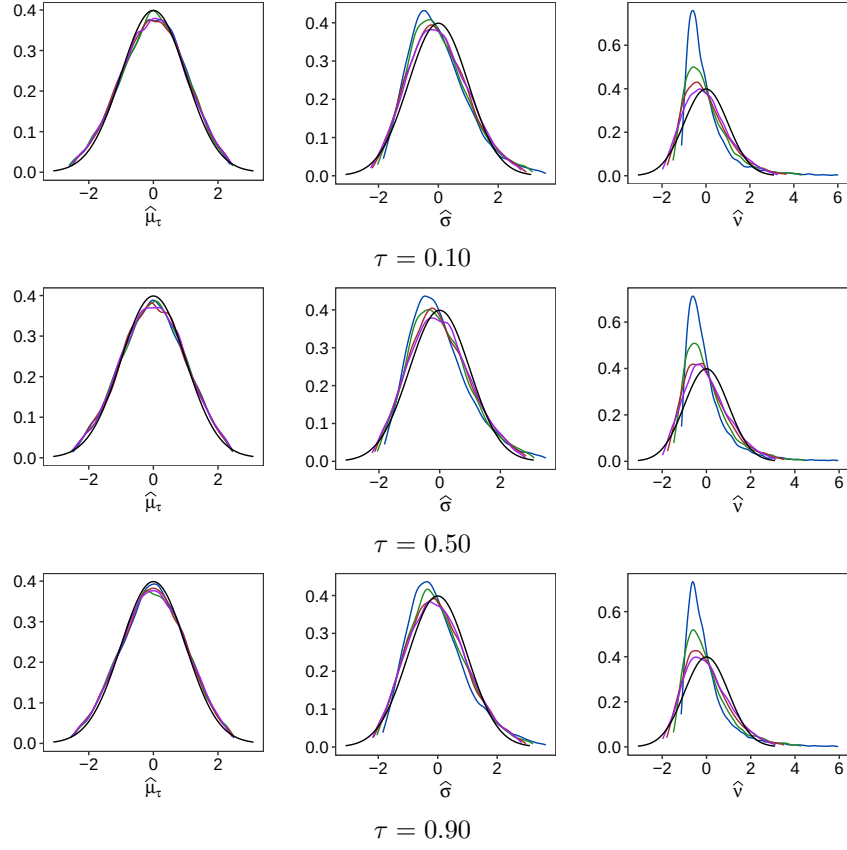


FIGURE 8: Sampling densities of the maximum likelihood estimators of the parameters of the $QGG(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.4. Scenario 4— $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$

TABLE 4: Monte Carlo simulation results for the QGG(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	1.9036	0.9913	0.5285	1.8607	0.9814	0.5124	1.8326	0.9993	0.5079
	200	1.8762	0.9687	0.5126	1.8506	0.9610	0.5068	1.8413	0.9762	0.5036
	400	1.8648	0.9552	0.5085	1.8510	0.9567	0.5033	1.8460	0.9626	0.5016
	800	1.8561	0.9535	0.5035	1.8516	0.9534	0.5013	1.8472	0.9555	0.5010
Bias	100	0.0536	0.0413	0.0285	0.0107	0.0314	0.0124	-0.0174	0.0493	0.0079
	200	0.0262	0.0187	0.0126	0.0006	0.0110	0.0068	-0.0087	0.0262	0.0036
	400	0.0148	0.0052	0.0085	0.0010	0.0067	0.0033	-0.0040	0.0126	0.0016
	800	0.0061	0.0035	0.0035	0.0016	0.0034	0.0013	-0.0028	0.0055	0.0010
SD	100	0.4152	0.2418	0.1717	0.1741	0.1960	0.0997	0.1204	0.2606	0.0836
	200	0.2967	0.1529	0.1068	0.1244	0.1323	0.0683	0.0856	0.1840	0.0575
	400	0.2098	0.1036	0.0737	0.0878	0.0933	0.0482	0.0605	0.1237	0.0400
	800	0.1500	0.0730	0.0505	0.1452	0.0655	0.0334	0.0425	0.0873	0.0282
MSE	100	0.1753	0.0602	0.0303	0.0304	0.0394	0.0101	0.0148	0.0703	0.0070
	200	0.0887	0.0237	0.0116	0.0155	0.0176	0.0047	0.0074	0.0346	0.0033
	400	0.0442	0.0108	0.0055	0.0077	0.0087	0.0023	0.0037	0.0155	0.0016
	800	0.0225	0.0053	0.0026	0.0211	0.0043	0.0011	0.0018	0.0077	0.0008
CP	100	0.9031	0.9188	0.9350	0.9389	0.9560	0.9530	0.9295	0.9548	0.9487
	200	0.9153	0.9363	0.9418	0.9406	0.9512	0.9518	0.9407	0.9444	0.9447
	400	0.9207	0.9371	0.9427	0.9418	0.9461	0.9444	0.9437	0.9457	0.9434
	800	0.9204	0.9343	0.9369	0.9453	0.9424	0.9469	0.9491	0.9442	0.9444

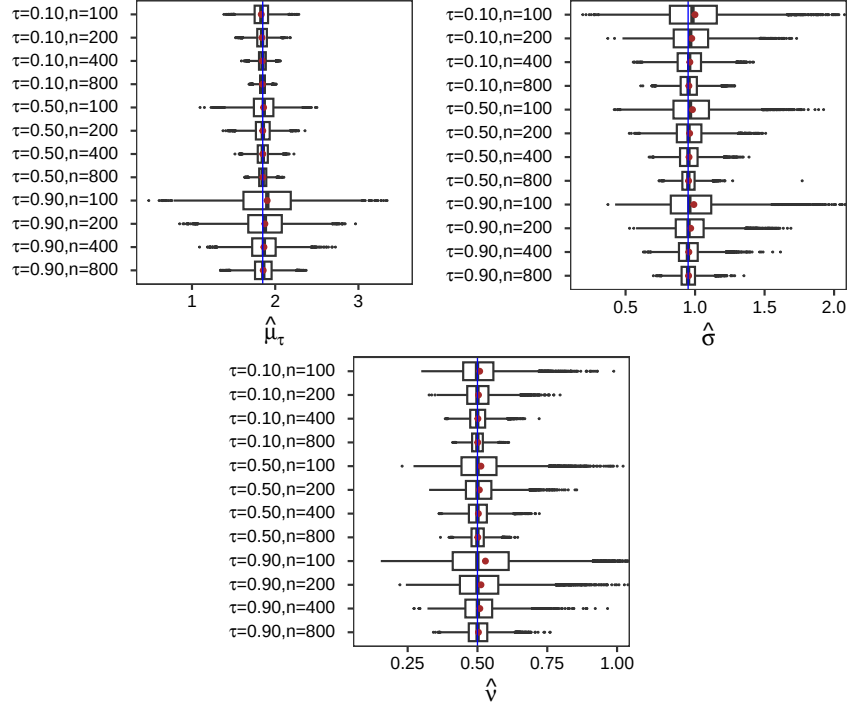


FIGURE 9: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $QGG(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$, considering $n \in \{100, 200, 400\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: $-$: true parameter value; $\bullet$: mean of the 10,000 replications.

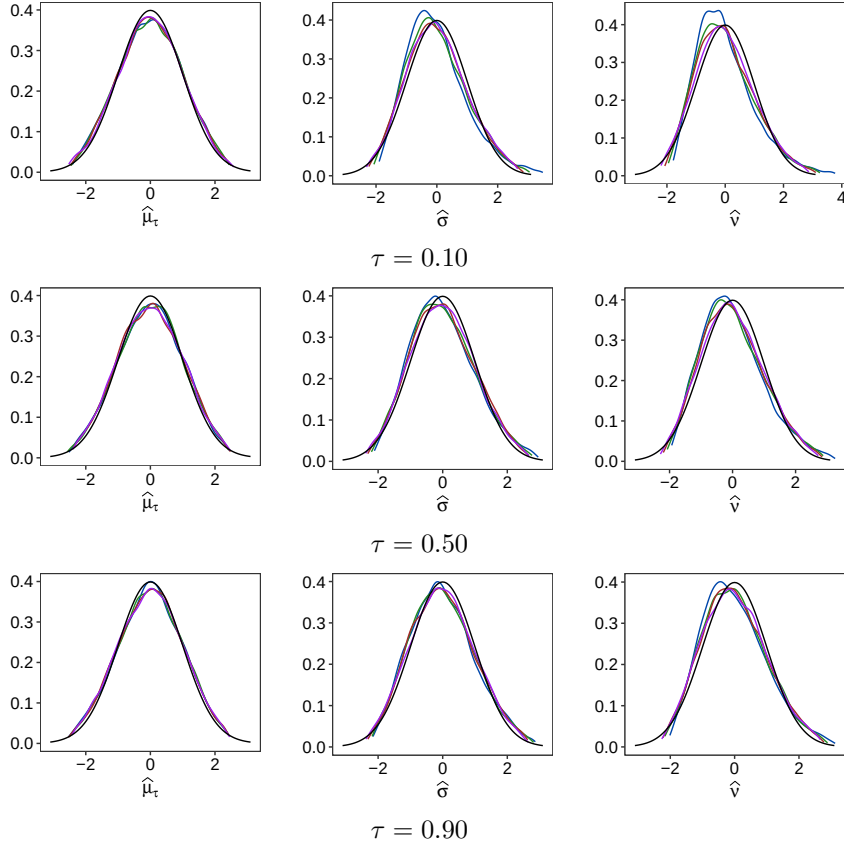


FIGURE 10: Sampling densities of the maximum likelihood estimators of the parameters of the $QGG(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4. QR-QGG regression

This section presents additional results from the Monte Carlo simulations for the QR-QGG regression model. The tables report the mean estimates, biases, standard deviations, mean squared errors, and coverage rates of the 95% confidence intervals for the scenarios, sample sizes, and quantile levels considered.

The plots complement the analysis by displaying boxplots and sampling densities of the maximum likelihood estimators, thereby showing the dispersion of the estimates, their concentration around the true values, and the asymptotic behavior of the estimators.

4.1. Scenario 1— $\sigma = 1.5$, $\nu = 0.5$, $\beta = (0.5, 0.3, 0.2)$

TABLE 5: Monte Carlo simulation results for the maximum likelihood estimators of the QR-QGG regression model with $\sigma = 1.5$, $\nu = 0.5$, $\beta_0 = 0.5$, $\beta_1 = 0.3$, $\beta_2 = 0.2$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5960	0.0960	0.5763	0.3414	0.9356	1.5419	0.0419	0.3586	0.1304	0.9457
	0.5	1.5580	0.0580	0.4354	0.1930	0.9410	1.5319	0.0319	0.2827	0.0809	0.9483
	0.9	1.5422	0.0422	0.3697	0.1385	0.9472	1.5210	0.0210	0.2432	0.0596	0.9525
$\hat{\nu}$	0.1	0.5300	0.0300	0.1323	0.0184	0.9636	0.5142	0.0142	0.0851	0.0074	0.9567
	0.5	0.5397	0.0397	0.1516	0.0246	0.9603	0.5172	0.0172	0.1015	0.0106	0.9505
	0.9	0.5595	0.0595	0.2137	0.0492	0.9524	0.5302	0.0302	0.1426	0.0212	0.9526
$\hat{\beta}_0$	0.1	0.5142	0.0142	0.1806	0.0328	0.9356	0.5033	0.0033	0.1306	0.0171	0.9421
	0.5	0.5042	0.0042	0.1319	0.0174	0.9428	0.4997	-0.0003	0.0990	0.0098	0.9479
	0.9	0.4835	-0.0165	0.1525	0.0235	0.9396	0.4926	-0.0074	0.1033	0.0107	0.9449
$\hat{\beta}_1$	0.1	0.2940	-0.0060	0.1879	0.0353	0.9482	0.3019	0.0019	0.1289	0.0166	0.9496
	0.5	0.2986	-0.0014	0.1777	0.0316	0.9425	0.2994	-0.0006	0.1292	0.0167	0.9460
	0.9	0.2973	-0.0027	0.1938	0.0376	0.9412	0.2982	-0.0018	0.1238	0.0153	0.9461
$\hat{\beta}_2$	0.1	0.1973	-0.0027	0.1827	0.0334	0.9442	0.2012	0.0012	0.1213	0.0147	0.9504
	0.5	0.1941	-0.0059	0.1741	0.0303	0.9424	0.2011	0.0011	0.1265	0.0160	0.9466
	0.9	0.2024	0.0024	0.1788	0.0320	0.9405	0.1987	-0.0013	0.1345	0.0181	0.9426
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5208	0.0208	0.2430	0.0595	0.9448	1.5122	0.0122	0.1646	0.0272	0.9492
	0.5	1.5173	0.0173	0.1964	0.0389	0.9456	1.5059	0.0059	0.1303	0.0170	0.9515
	0.9	1.5093	0.0093	0.1673	0.0281	0.9474	1.5049	0.0049	0.1163	0.0135	0.9522
$\hat{\nu}$	0.1	0.5066	0.0066	0.0583	0.0034	0.9538	0.5031	0.0031	0.0403	0.0016	0.9556
	0.5	0.5081	0.0081	0.0702	0.0050	0.9473	0.5044	0.0044	0.0474	0.0023	0.9536
	0.9	0.5156	0.0156	0.0981	0.0099	0.9504	0.5070	0.0070	0.0672	0.0046	0.9529
$\hat{\beta}_0$	0.1	0.5012	0.0012	0.0909	0.0083	0.9498	0.5003	0.0003	0.0648	0.0042	0.9515
	0.5	0.5007	0.0007	0.0703	0.0049	0.9460	0.5000	0.0000	0.0510	0.0026	0.9523
	0.9	0.4955	-0.0045	0.0701	0.0049	0.9499	0.4988	-0.0012	0.0501	0.0025	0.9493
$\hat{\beta}_1$	0.1	0.3008	0.0008	0.0910	0.0083	0.9548	0.3003	0.0003	0.0639	0.0041	0.9488
	0.5	0.2994	-0.0006	0.0899	0.0081	0.9505	0.3001	0.0001	0.0621	0.0039	0.9507
	0.9	0.2994	-0.0006	0.0884	0.0078	0.9508	0.2987	-0.0013	0.0632	0.0040	0.9522
$\hat{\beta}_2$	0.1	0.2008	0.0008	0.0889	0.0079	0.9515	0.2022	0.0022	0.0643	0.0041	0.9530
	0.5	0.1996	-0.0004	0.0895	0.0080	0.9514	0.1994	-0.0006	0.0630	0.0040	0.9503
	0.9	0.2012	0.0012	0.0882	0.0078	0.9470	0.1998	-0.0002	0.0649	0.0042	0.9508

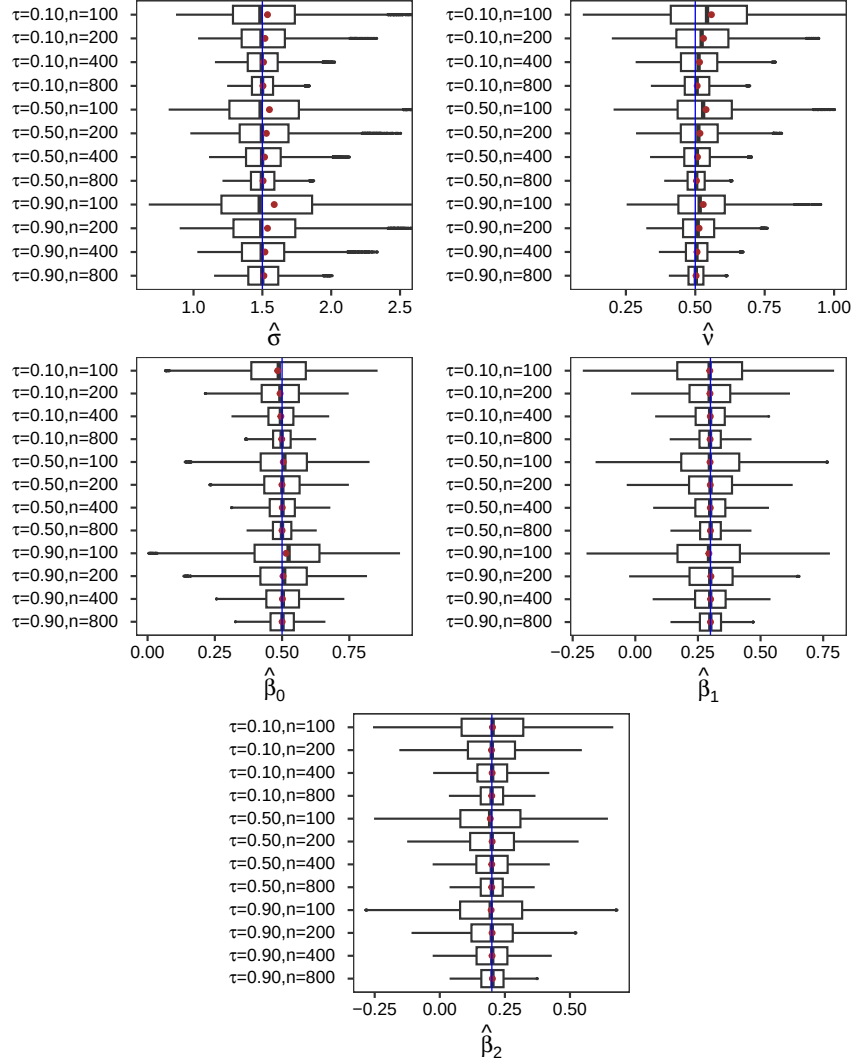


FIGURE 11: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the QR-QGG regression model with $\sigma = 1.5$, $\nu = 0.5$, $\beta_0 = 0.5$, $\beta_1 = 0.3$ and $\beta_2 = 0.2$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: $-$: true parameter value; $\bullet$: mean of the 10,000 replications.

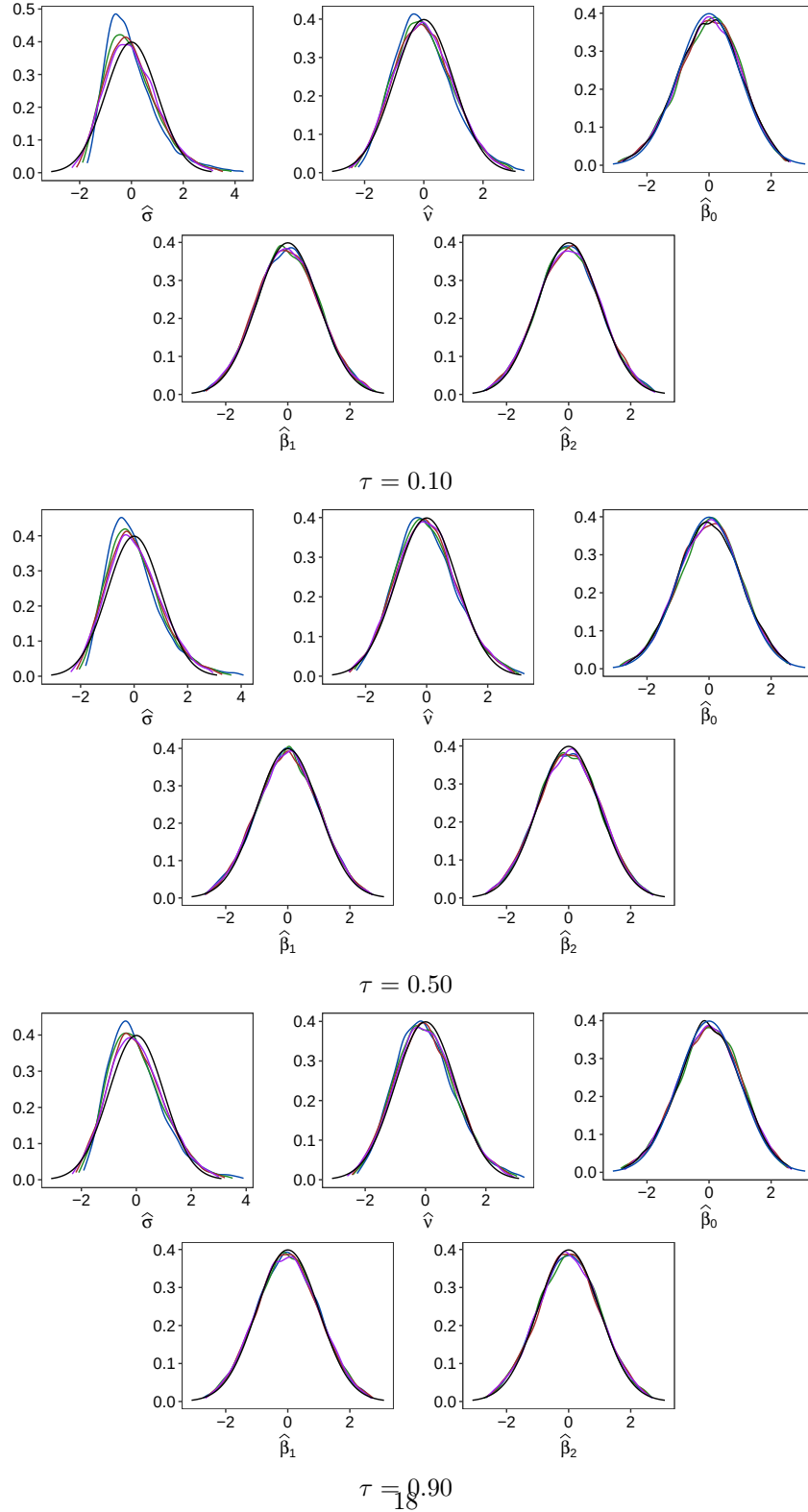


FIGURE 12: Sampling densities of the maximum likelihood estimators of the parameters of the QR-QGG regression model obtained via Monte Carlo simulation, considering $\sigma = 1.5$, $\nu = 0.5$, $\beta = (0.5, 0.3, 0.2)$, $\tau \in \{0.10, 0.50, 0.90\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4.2. Scenario 3— $\sigma = 1.5$, $\nu = 2$, $\beta = (0.3, 0.5, 0.2)$

TABLE 6: Monte Carlo simulation results for the maximum likelihood estimators of the QR-QGG regression model with $\sigma = 1.5$, $\nu = 2.0$, $\beta_0 = 0.3$, $\beta_1 = 0.5$, $\beta_2 = 0.2$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.6790	0.1790	1.0311	1.0952	0.8811	1.5833	0.0833	0.5822	0.3459	0.9252
	0.5	1.6598	0.1598	0.8550	0.7566	0.9169	1.5668	0.0668	0.5029	0.2574	0.9364
	0.9	1.6205	0.1205	0.6938	0.4959	0.9299	1.5479	0.0479	0.4067	0.1677	0.9424
$\hat{\nu}$	0.1	2.2390	0.2390	0.7647	0.6419	0.9551	2.0863	0.0863	0.4111	0.1764	0.9513
	0.5	2.1627	0.1627	0.6419	0.4385	0.9550	2.0720	0.0720	0.3718	0.1434	0.9544
	0.9	2.1293	0.1293	0.5587	0.3289	0.9586	2.0630	0.0630	0.3410	0.1203	0.9611
$\hat{\beta}_0$	0.1	0.3060	0.0060	0.0841	0.0071	0.9349	0.3043	0.0043	0.0578	0.0034	0.9388
	0.5	0.3025	0.0025	0.0656	0.0043	0.9417	0.3003	0.0003	0.0451	0.0020	0.9474
	0.9	0.2925	-0.0075	0.0625	0.0040	0.9417	0.2962	-0.0038	0.0431	0.0019	0.9479
$\hat{\beta}_1$	0.1	0.4968	-0.0032	0.0750	0.0056	0.9352	0.4981	-0.0019	0.0550	0.0030	0.9397
	0.5	0.4961	-0.0039	0.0793	0.0063	0.9409	0.4990	-0.0010	0.0537	0.0029	0.9467
	0.9	0.5006	0.0006	0.0739	0.0055	0.9415	0.5005	0.0005	0.0522	0.0027	0.9476
$\hat{\beta}_2$	0.1	0.1989	-0.0011	0.0762	0.0058	0.9333	0.1979	-0.0021	0.0511	0.0026	0.9389
	0.5	0.1985	-0.0015	0.0775	0.0060	0.9456	0.2003	0.0003	0.0515	0.0027	0.9470
	0.9	0.2018	0.0018	0.0733	0.0054	0.9462	0.2000	0.0000	0.0495	0.0024	0.9480
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5365	0.0365	0.3674	0.1363	0.9360	1.5177	0.0177	0.2458	0.0607	0.9428
	0.5	1.5299	0.0299	0.3170	0.1014	0.9412	1.5122	0.0122	0.2130	0.0455	0.9452
	0.9	1.5213	0.0213	0.2692	0.0729	0.9491	1.5138	0.0138	0.1847	0.0343	0.9525
$\hat{\nu}$	0.1	2.0405	0.0405	0.2665	0.0727	0.9514	2.0185	0.0185	0.1813	0.0332	0.9508
	0.5	2.0330	0.0330	0.2468	0.0620	0.9493	2.0172	0.0172	0.1681	0.0286	0.9532
	0.9	2.0308	0.0308	0.2304	0.0540	0.9550	2.0115	0.0115	0.1599	0.0257	0.9513
$\hat{\beta}_0$	0.1	0.3018	0.0018	0.0418	0.0018	0.9440	0.3008	0.0008	0.0292	0.0009	0.9475
	0.5	0.2999	-0.0001	0.0321	0.0010	0.9514	0.3001	0.0001	0.0217	0.0005	0.9515
	0.9	0.2982	-0.0018	0.0310	0.0010	0.9528	0.2992	-0.0008	0.0218	0.0005	0.9514
$\hat{\beta}_1$	0.1	0.4986	-0.0014	0.0388	0.0015	0.9475	0.4997	-0.0003	0.0277	0.0008	0.9452
	0.5	0.4999	-0.0001	0.0391	0.0015	0.9480	0.4999	-0.0001	0.0257	0.0007	0.9544
	0.9	0.5001	0.0001	0.0359	0.0013	0.9508	0.4998	-0.0002	0.0252	0.0006	0.9532
$\hat{\beta}_2$	0.1	0.1998	-0.0002	0.0373	0.0014	0.9433	0.1996	-0.0004	0.0251	0.0006	0.9502
	0.5	0.2000	0.0000	0.0360	0.0013	0.9512	0.1999	-0.0001	0.0262	0.0007	0.9506
	0.9	0.2002	0.0002	0.0377	0.0014	0.9536	0.2000	0.0000	0.0259	0.0007	0.9511

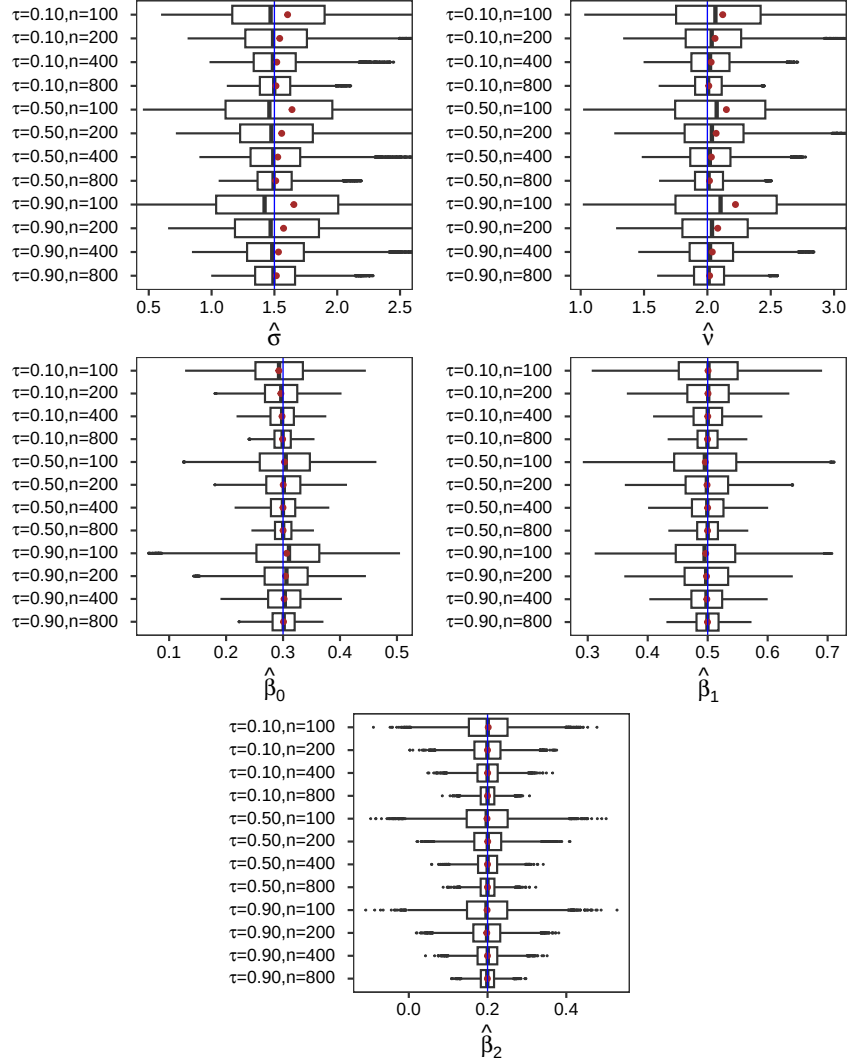


FIGURE 13: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the QR-QGG regression model with $\sigma = 1.5$, $\nu = 2$, $\beta_0 = 0.3$, $\beta_1 = 0.5$ and $\beta_2 = 0.2$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

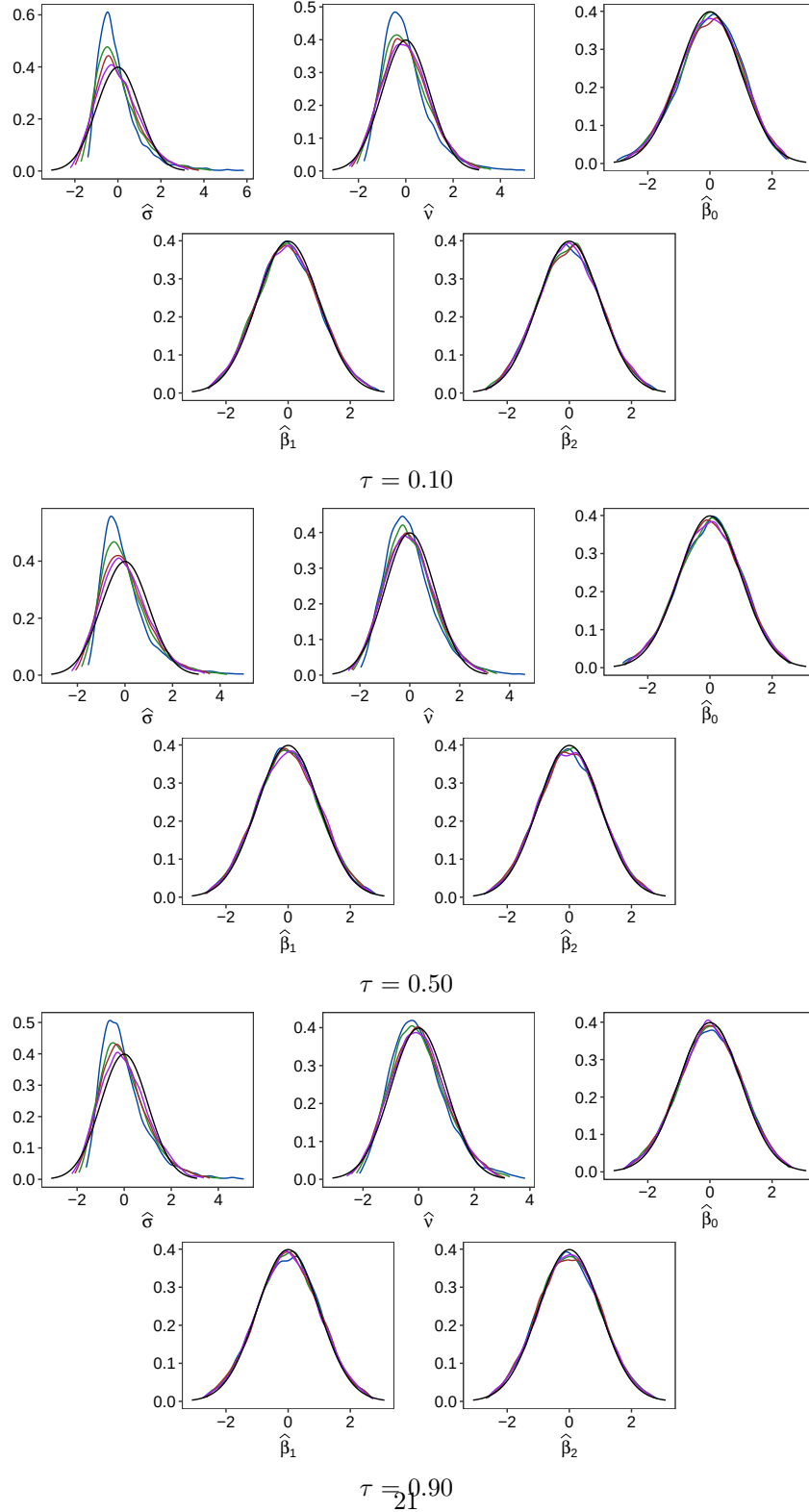


FIGURE 14: Sampling densities of the maximum likelihood estimators of the parameters of the $QGG(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\sigma = 1.5$, $\nu = 2$, $\beta_0 = 0.3$, $\beta_1 = 0.5$ and $\beta_2 = 0.2$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4.3. Scenario 6— $\sigma = 1.0$, $\nu = 0.4$, $\beta = (1.1, -0.9, 2.0)$

TABLE 7: Monte Carlo simulation results for the maximum likelihood estimators of the QR-QGG regression model with $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$, $\beta_2 = 2.0$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.0574	0.0574	0.3947	0.1591	0.9264	1.0230	0.0230	0.2294	0.0532	0.9407
	0.5	1.0269	0.0269	0.2704	0.0739	0.9457	1.0153	0.0153	0.1866	0.0350	0.9483
	0.9	1.0260	0.0260	0.2246	0.0511	0.9497	1.0115	0.0115	0.1523	0.0233	0.9512
$\hat{\nu}$	0.1	0.4267	0.0267	0.1059	0.0119	0.9568	0.4115	0.0115	0.0608	0.0038	0.9527
	0.5	0.4268	0.0268	0.0926	0.0093	0.9590	0.4120	0.0120	0.0597	0.0037	0.9530
	0.9	0.4278	0.0278	0.0932	0.0095	0.9634	0.4135	0.0135	0.0614	0.0040	0.9569
$\hat{\beta}_0$	0.1	1.1137	0.0137	0.1880	0.0355	0.9293	1.1095	0.0095	0.1250	0.0157	0.9405
	0.5	1.1078	0.0078	0.1148	0.0132	0.9391	1.1025	0.0025	0.0842	0.0071	0.9449
	0.9	1.0864	-0.0136	0.1233	0.0154	0.9439	1.0950	-0.0050	0.0795	0.0063	0.9480
$\hat{\beta}_1$	0.1	-0.8919	0.0081	0.1395	0.0195	0.9349	-0.8964	0.0036	0.0920	0.0085	0.9436
	0.5	-0.8928	0.0072	0.1205	0.0146	0.9411	-0.8974	0.0026	0.0810	0.0066	0.9454
	0.9	-0.8943	0.0057	0.1373	0.0189	0.9432	-0.9004	-0.0004	0.0797	0.0064	0.9484
$\hat{\beta}_2$	0.1	1.9822	-0.0178	0.2185	0.0481	0.9302	1.9881	-0.0119	0.1323	0.0176	0.9417
	0.5	1.9811	-0.0189	0.1333	0.0181	0.9404	1.9945	-0.0055	0.0970	0.0094	0.9455
	0.9	1.9988	-0.0012	0.1464	0.0214	0.9432	1.9983	-0.0017	0.0980	0.0096	0.9472
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.0142	0.0142	0.1553	0.0243	0.9466	1.0058	0.0058	0.1055	0.0112	0.9456
	0.5	1.0080	0.0080	0.1233	0.0153	0.9491	1.0032	0.0032	0.0863	0.0075	0.9514
	0.9	1.0052	0.0052	0.1040	0.0108	0.9494	1.0029	0.0029	0.0717	0.0051	0.9538
$\hat{\nu}$	0.1	0.4050	0.0050	0.0410	0.0017	0.9511	0.4026	0.0026	0.0283	0.0008	0.9455
	0.5	0.4051	0.0051	0.0406	0.0017	0.9497	0.4030	0.0030	0.0280	0.0008	0.9528
	0.9	0.4067	0.0067	0.0420	0.0018	0.9492	0.4029	0.0029	0.0291	0.0009	0.9495
$\hat{\beta}_0$	0.1	1.1061	0.0061	0.0874	0.0077	0.9436	1.1013	0.0013	0.0641	0.0041	0.9448
	0.5	1.1020	0.0020	0.0589	0.0035	0.9517	1.1003	0.0003	0.0418	0.0018	0.9475
	0.9	1.0976	-0.0024	0.0590	0.0035	0.9499	1.0989	-0.0011	0.0393	0.0015	0.9526
$\hat{\beta}_1$	0.1	-0.8982	0.0018	0.0644	0.0041	0.9486	-0.8986	0.0014	0.0459	0.0021	0.9503
	0.5	-0.8990	0.0010	0.0583	0.0034	0.9495	-0.8998	0.0002	0.0412	0.0017	0.9518
	0.9	-0.8999	0.0001	0.0581	0.0034	0.9481	-0.9002	-0.0002	0.0407	0.0017	0.9529
$\hat{\beta}_2$	0.1	1.9935	-0.0065	0.0937	0.0088	0.9448	1.9983	-0.0017	0.0676	0.0046	0.9483
	0.5	1.9965	-0.0035	0.0667	0.0045	0.9521	1.9990	-0.0010	0.0471	0.0022	0.9493
	0.9	1.9996	-0.0004	0.0661	0.0044	0.9492	1.9998	-0.0002	0.0455	0.0021	0.9553

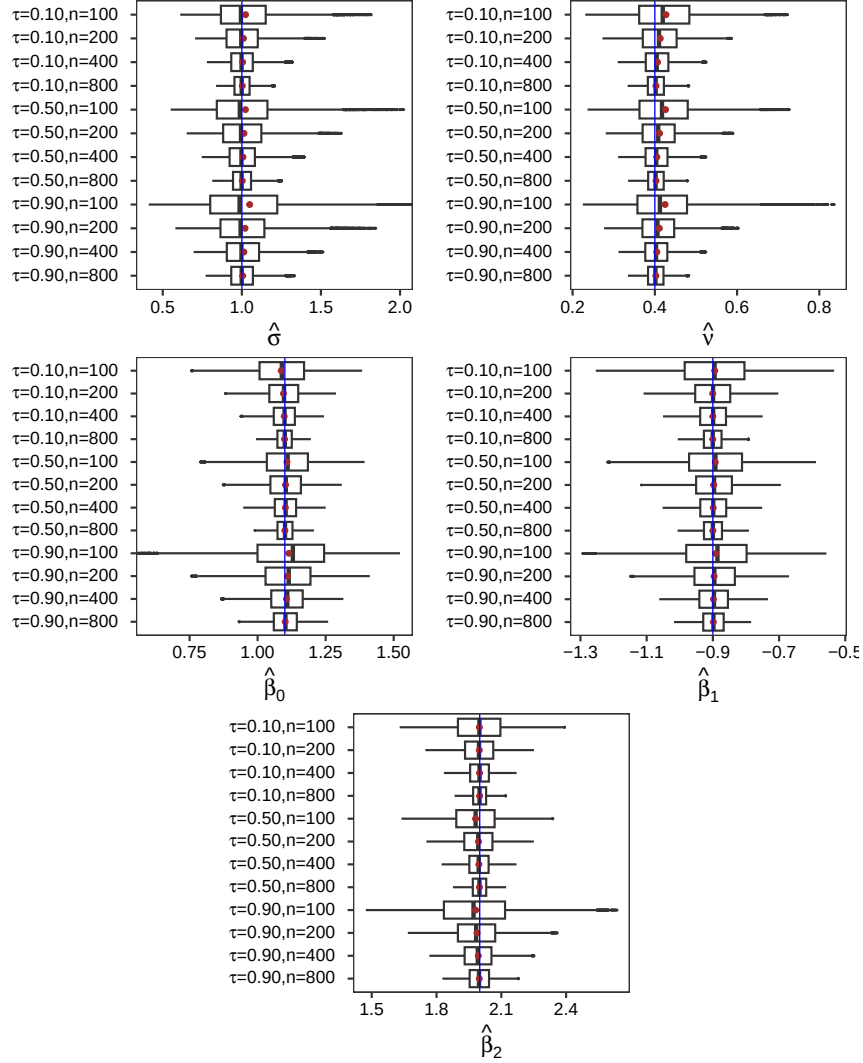


FIGURE 15: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the QR-QGG regression model with $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$ and $\beta_2 = 2.0$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: $-$: true parameter value; $\bullet$: mean of the 10,000 replications.

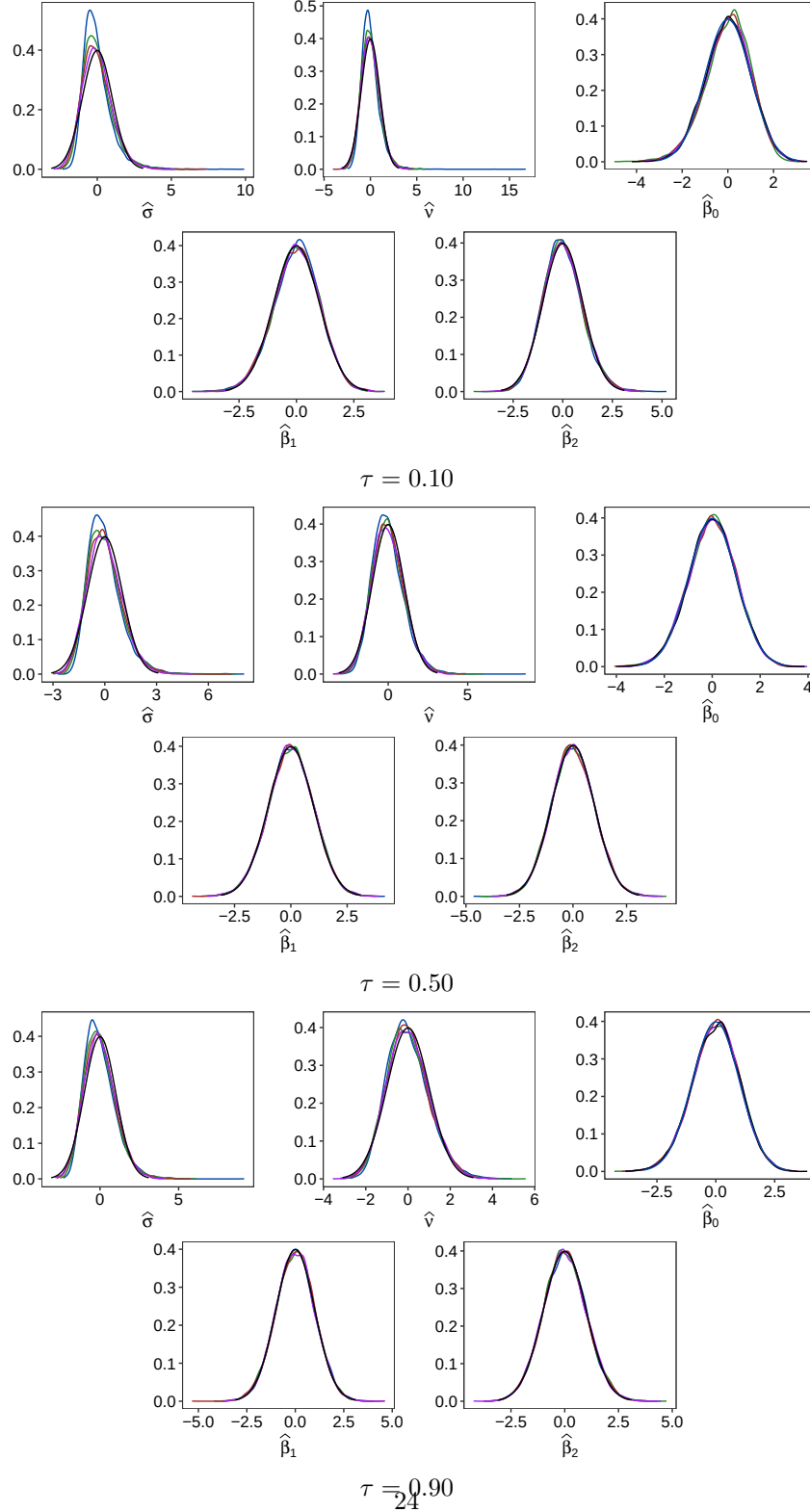


FIGURE 16: Sampling densities of the maximum likelihood estimators of the parameters of the QR-QGG regression model obtained via Monte Carlo simulation, considering $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$ and $\beta_2 = 2.0$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.